Calculus II

Lecture 04

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Fall 2024

Integration by Substitution (1/7)

Definition (1/7):

If $u = f(x)$, the differential of u, written du, is defined as

$$du = f'(x)\,dx$$

For example, if $u = 2x^3 + 1$, then $du = 6x^2\,dx$.

Integration by Substitution (1/7)

Definition (2/7):

Find

$$\int (2x^3 + 1)^4 6x^2 \, dx.$$

Integration by Substitution (1/7)

Definition (3/7):

Find

$$\int \underline{(2x^3 + 1)}^4 6x^2\,dx.$$

Let $u = 2x^3 + 1$; then $du = 6x^2\,dx$.

Integration by Substitution (1/7)

Definition (4/7):

Let $u = 2x^3 + 1$; then $du = 6x^2\,dx$.

$$\int (2x^3 + 1)^4 6x^2\,dx = \int (\overbrace{2x^3 + 1}^{u})^4 (\overbrace{6x^2\,dx}^{du})$$

$$= \int u^4 du$$

Integration by Substitution (1/7)

Definition (5/7):

Let $u = 2x^3 + 1$; then $du = 6x^2\,dx$.

$$\int u^4\,du = \frac{u^5}{5} + C$$

Integration by Substitution (1/7)

Definition (6/7):

Let $u = 2x^3 + 1$; then $du = 6x^2\,dx$.

$$\int u^4\,du = \frac{u^5}{5} + C$$

$$= \frac{(2x^3 + 1)^5}{5} + C.$$

Integration by Substitution (1/7)

Definition (7/7):

Chan Rule

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x).$$

$$\frac{d}{dx}\left[\frac{(2x^3+1)^5}{5} + C\right] = \frac{1}{5} \cdot 5(2x^3+1)^4(6x^2) + 0$$

$$= (2x^3+1)^4 6x^2.$$

Integration by Substitution (2/7)

GUIDELINES

1. Choose a substitution $u = g(x)$. Usually, it is best to choose the *inner* part of a composite function, such as a quantity raised to a power.
2. Compute $du = g'(x)\,dx$.
3. Rewrite the integral in terms of the variable u.
4. Find the resulting integral in terms of u.
5. Replace u by $g(x)$ to obtain an antiderivative in terms of x.
6. Check your answer by differentiating.

Integration by Substitution (3/7)

Example 1 (1/4):

Find $$\int 2x\sqrt{1 + x^2}\, dx$$

Integration by Substitution (3/7)

Example 1 (2/4):

$$\int 2x\sqrt{1+x^2}\,dx$$

Let $u = 1 + x^2$, → $du = 2x\,dx$

Integration by Substitution (3/7)

Example 1 (3/4):

$$\int 2x\sqrt{1+x^2}\,dx$$

Let $u = 1 + x^2$, → $du = 2x\,dx$

$$= \int \sqrt{1+x^2}\,2x\,dx = \int \sqrt{u}\,du$$

$$= \tfrac{2}{3}u^{3/2} + C = \tfrac{2}{3}(1+x^2)^{3/2} + C$$

Integration by Substitution (3/7)

Example 1 (4/4):

$$\int 2x\sqrt{1+x^2}\,dx = \tfrac{2}{3}(1+x^2)^{3/2} + C$$

$$\frac{d}{dx}\left[\tfrac{2}{3}(1+x^2)^{3/2} + C\right] =$$

$$= \frac{2}{3}\cdot\frac{3}{2}(1+x^2)^{1/2}\cdot 2x$$

$$= 2x\sqrt{1+x^2}$$

Integration by Substitution (3/7)

Example 1 (4/4):

$$\int \boxed{2x\sqrt{1+x^2}}\,dx = \frac{2}{3}(1+x^2)^{3/2} + C$$

$$\frac{d}{dx}\left[\frac{2}{3}(1+x^2)^{3/2} + C\right] =$$

$$= \frac{2}{3} \cdot \frac{3}{2}(1+x^2)^{1/2} \cdot 2x$$

$$= \boxed{2x\sqrt{1+x^2}}$$

Integration by Substitution (4/7)

Example 2 (1/3):

Evaluate $\int \sqrt{2x + 1}\, dx$.

Integration by Substitution (4/7)

Example 2 (2/3):

$$\int \sqrt{2x+1}\,dx.$$

Let $u = 2x + 1, \quad \rightarrow \quad du = 2\,dx$

Integration by Substitution (4/7)

Example 2 (3/3):

$\boxed{\textbf{Let } u = 2x + 1, \quad \rightarrow \quad du = 2\,dx}$

$$\int \sqrt{2x+1}\,dx = \int \sqrt{u} \cdot \tfrac{1}{2}\,du = \tfrac{1}{2}\int u^{1/2}\,du$$

$$= \frac{1}{2}\cdot\frac{u^{3/2}}{3/2} + C = \tfrac{1}{3}u^{3/2} + C$$

$$= \tfrac{1}{3}(2x+1)^{3/2} + C$$

Integration by Substitution (5/7)

Example 3 (1/4):

Find

$$\int \cos^2 x \sin x \, dx$$

Integration by Substitution (5/7)

Example 3 (2/4):

$$\int \cos^2 x \sin x \, dx$$

$$\textbf{Let } u = \cos x, \quad \rightarrow \quad du = -\sin x \, dx$$

Integration by Substitution (5/7)

Example 3 (3/4):

$$\int \cos^2 x \sin x \, dx$$

$$= -\int \overbrace{(\cos x)^2}^{u^2} \overbrace{(-\sin x)\, dx}^{du}$$

Integration by Substitution (5/7)

Example 3 (4/4):

$$= -\int \overbrace{(\cos x)^2}^{u^2} \overbrace{(-\sin x)\, dx}^{du}$$

$$= -\overbrace{\frac{(\cos x)^3}{3}}^{u^3/3} + C$$

Integration by Substitution (6/7)

Example 5 (1/3):

Find $\int x^2 e^{x^3}\, dx.$

Integration by Substitution (6/7)

Example 5 (2/3):

$$\int x^2 e^{x^3}\,dx$$

Let $u = x^3$, $\rightarrow$ $du = 3x^2\,dx$

$$\int x^2 e^{x^3}\,dx = \frac{1}{3}\int e^{x^3}(3x^2\,dx) = \frac{1}{3}\int e^u\,du$$

Integration by Substitution (6/7)

Example 5 (3/3):

Let $u = x^3$, → $du = 3x^2\,dx$

$$\int x^2 e^{x^3}\,dx = \frac{1}{3}\int e^{x^3}(3x^2\,dx) = \frac{1}{3}\int e^u\,du$$

$$= \frac{1}{3}e^u + C = \frac{1}{3}e^{x^3} + C.$$

Integration by Substitution (7/7)

Example 6 (1/6):

Calculate $\int x\sqrt{x+2}\,dx$

Integration by Substitution (7/7)

Example 6 (2/6):

$$\int x\sqrt{x+2}\,dx$$

$$\boxed{\textbf{Let } u = x + 2, \quad \rightarrow \quad du = dx}$$

Integration by Substitution (7/7)

Example 6 (3/6):

$$\int x\sqrt{x+2}\,dx$$

Let $u = x + 2, \quad \rightarrow \quad du = dx$

$x = u - 2$

Integration by Substitution (7/7)

Example 6 (4/6):

$$\int x\sqrt{x+2}\,dx$$

Let $u = x + 2, \quad \rightarrow \quad du = dx$

$$x = u - 2$$

$$= \int (u-2)\sqrt{u}\,du$$

Integration by Substitution (7/7)

Example 6 (5/6):

Let $u = x + 2, \quad \rightarrow \quad du = dx$

$$\int x\sqrt{x+2}\,dx$$

$$\int (u-2)\sqrt{u}\,du = \int u^{\frac{3}{2}} - 2u^{\left(\frac{1}{2}\right)}\,du$$

$$= \frac{2}{5}u^{\frac{5}{2}} - \frac{4}{3}u^{\frac{3}{2}} + C$$

Integration by Substitution (7/7)

Example 6 (6/6):

Let $u = x + 2$, → $du = dx$

$$\int x\sqrt{x+2}\,dx$$

$$= \frac{2}{5}u^{\frac{5}{2}} - \frac{4}{3}u^{\frac{3}{2}} + C$$

$$= \frac{2}{5}(x+2)^{\frac{5}{2}} - \frac{4}{3}(x+2)^{\frac{3}{2}} + C$$

Thank You

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